Centre for Biomedical Technologies

UNDERSTANDING THE RELIABILITY AND CONSISTENCY OF ULTRASOUND DERIVED ELASTOGRAPHY MEASURES OF THE HUMAN PLACENTA

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Centre for Biomedical Technologies

Queensland family COHORT STUDY

mater research

@edchris_
#QLDFCS

Supervisors

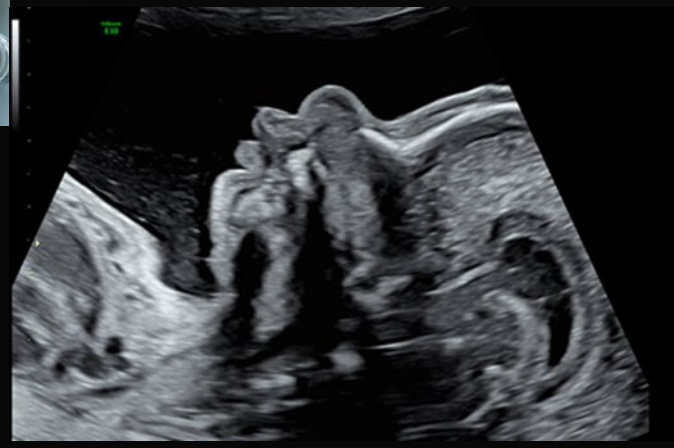
A/Professor Davide Fontanarosa

Dr Marie-Luise Wille

Professor Sailesh Kumar

QLD Family Cohort

- Pilot study to test the feasibility of a 10,000-family cohort
- Assess the health of newborns and their parents over three decades
- Broad range of qualitative and quantitative data collected



Overview

- Background
  - Placenta
  - Technology
- Aims & Objectives
- Methods
- Results
  - Thesis Overview – Chapters
- Discussion
  - Strength / Limitations
  - Future Recommendations

Placentation

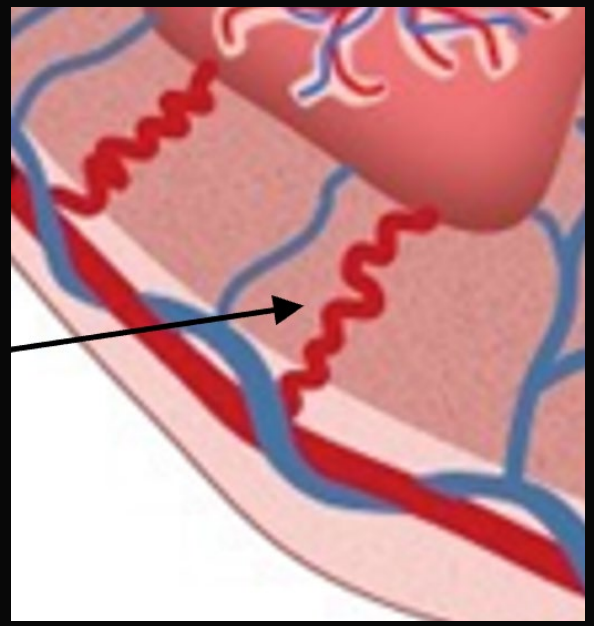
Cellular

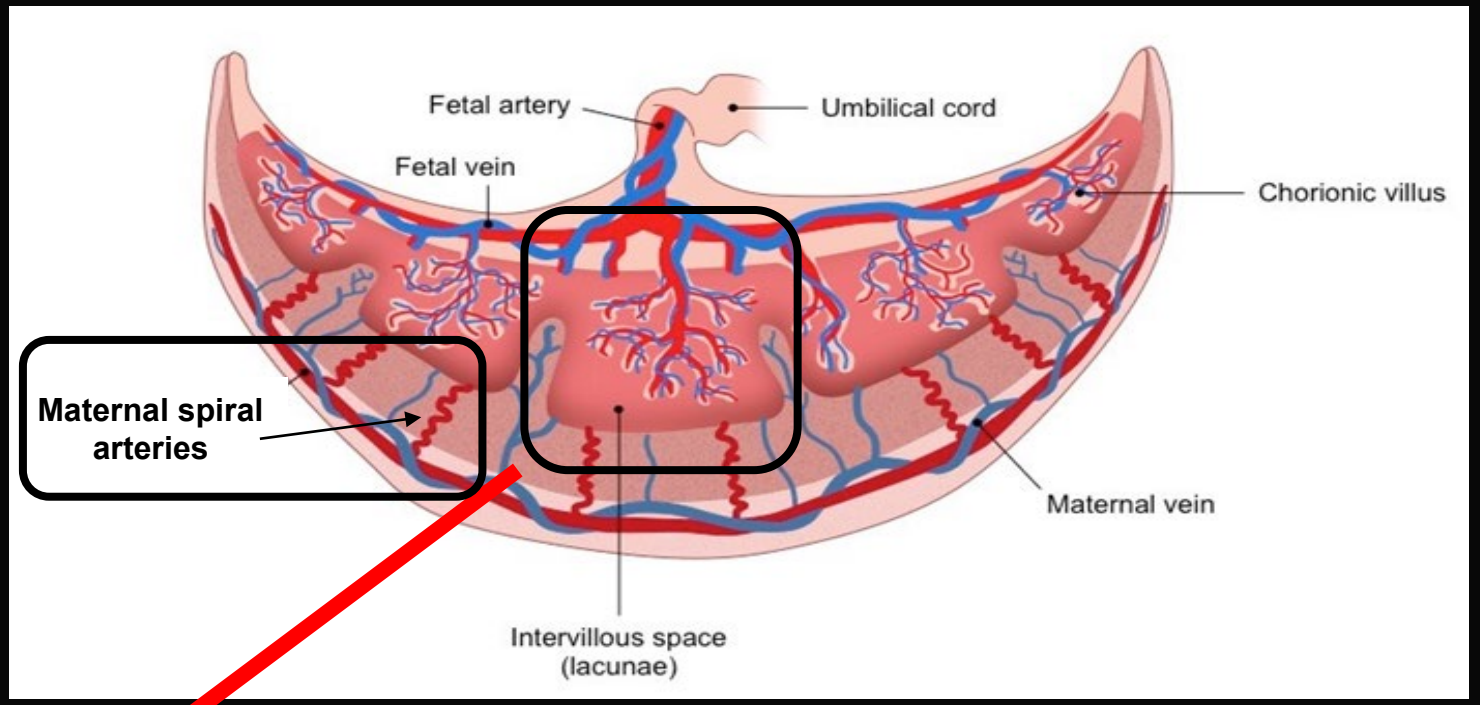
Maternal cardiovascular

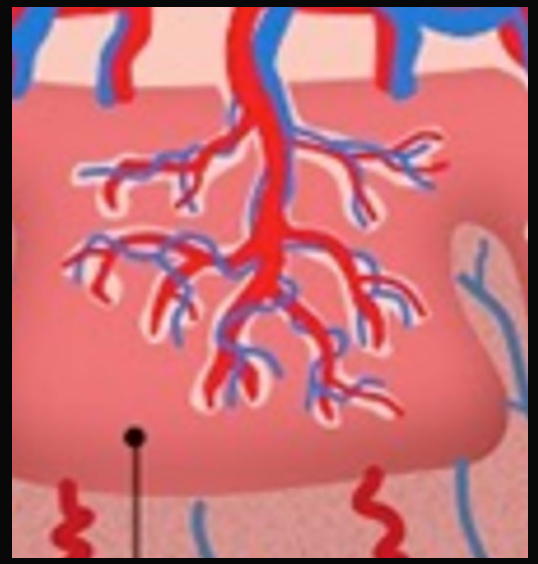
Growth Restriction

Preeclampsia

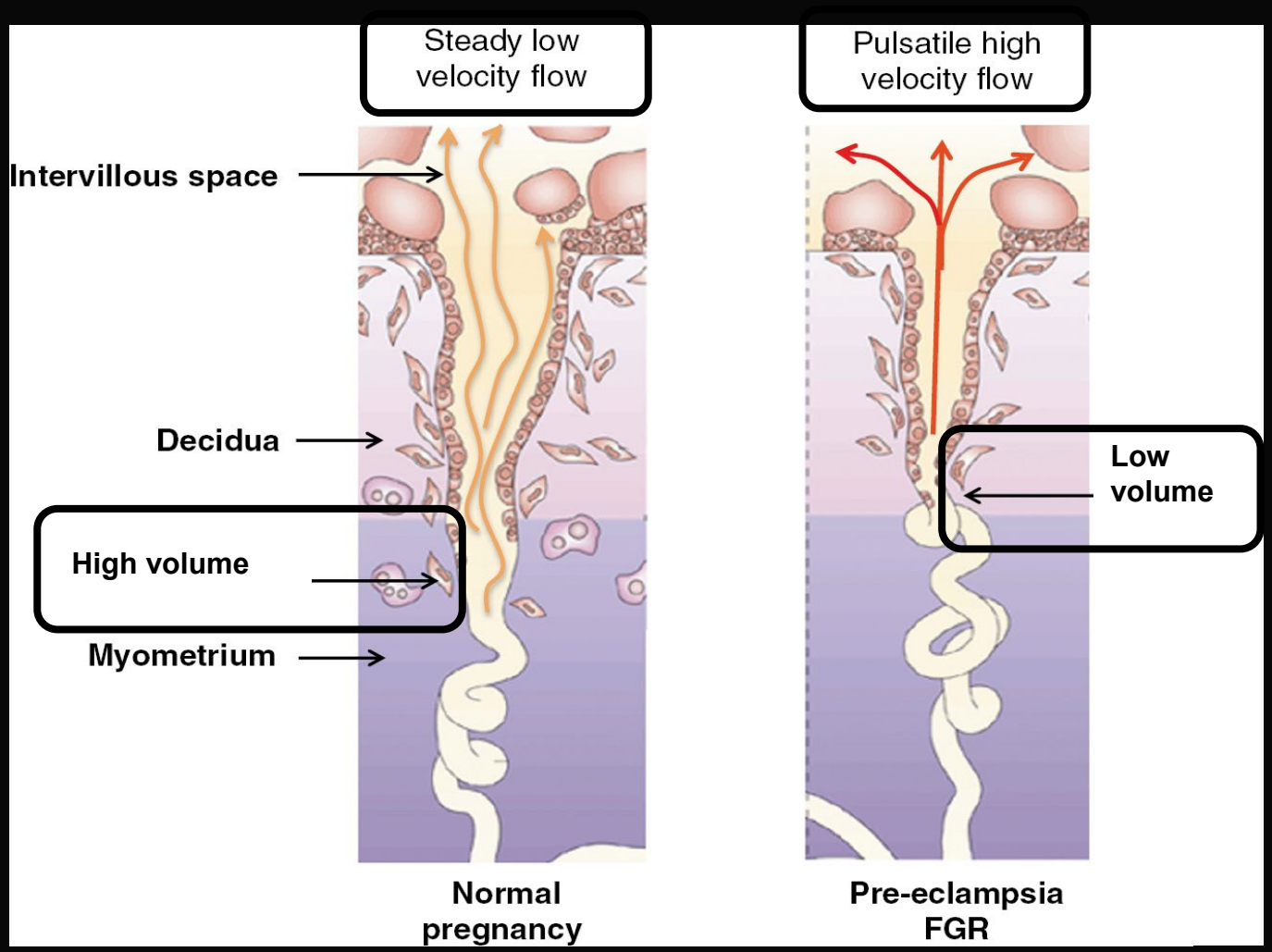
Gestational Diabetes

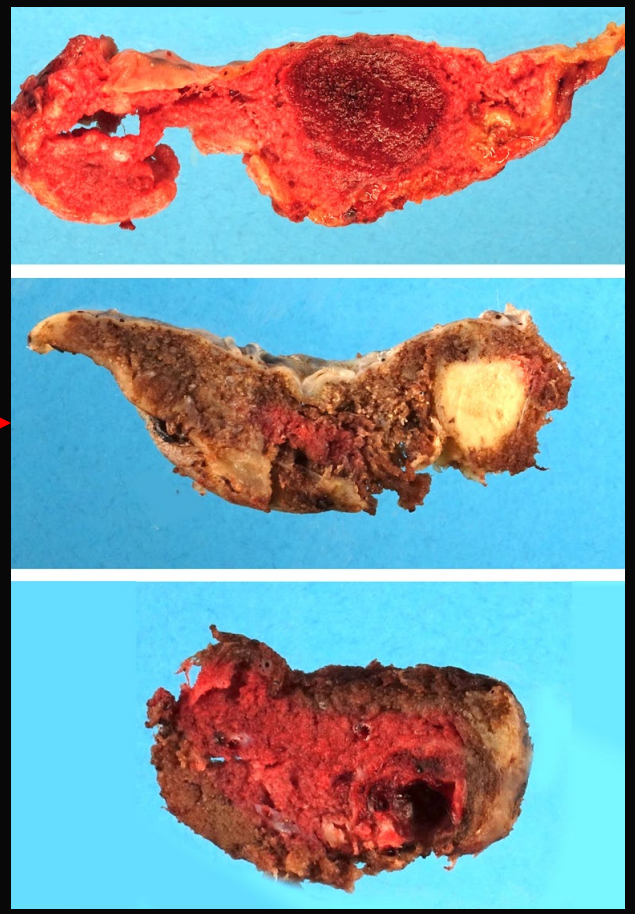




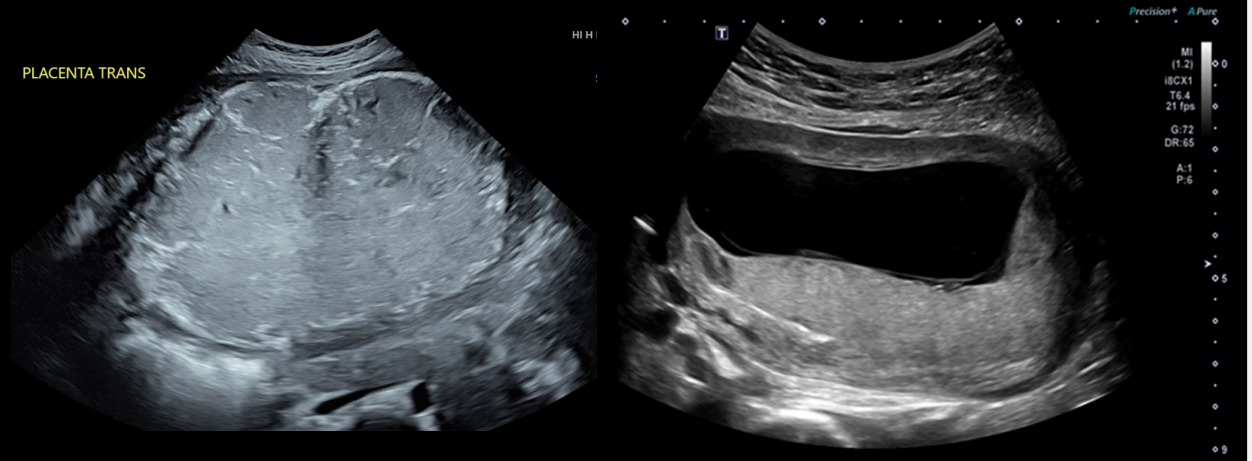


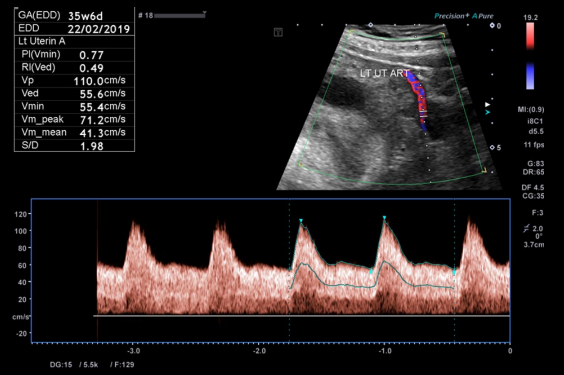
Cornell, B. 2016. Placenta. [ONLINE] Available at: http://ib.bioninja.com.au. [Accessed 15 Jan 2022]





Lees, C., Visser, G., & Hecher, K. (2018). Basic Principles. In Placental-Fetal Growth Restriction (pp. Xvi-63). Cambridge: Cambridge University Press.



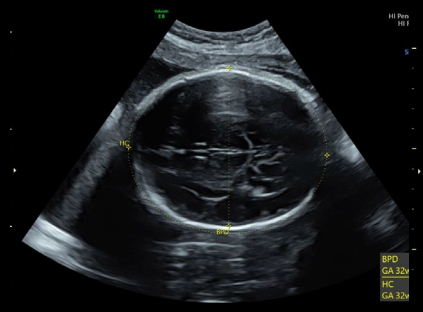


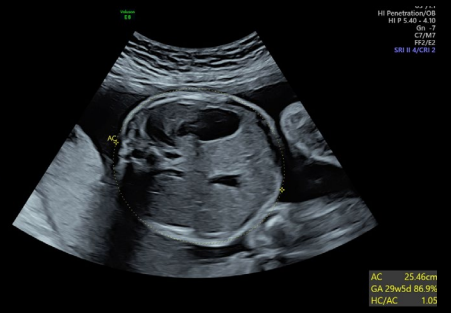
Anatomical

Haemodynamics

Placental Dysfunction

Biometric





Fetal Growth Restriction (FGR)

Early FGR

Prior to 32-34 weeks
Severe placental disease
Abnormal Dopplers

High morbidity and mortality

Detectable via ultrasound

Late FGR

After 34 weeks
Mild placental disease
Dopplers often normal

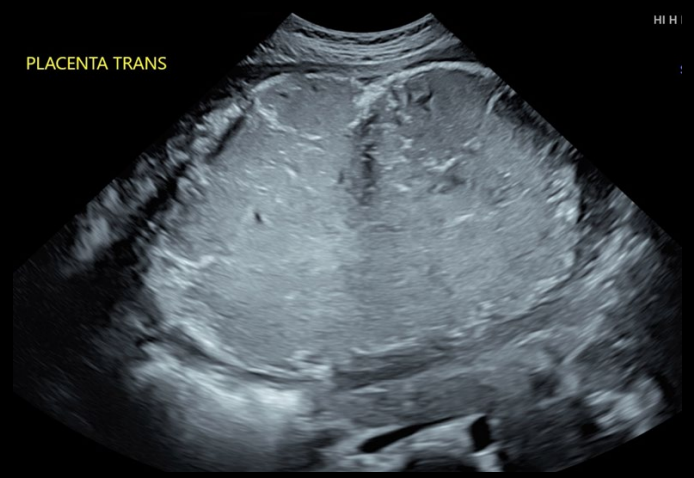
Lower morbidity

Often missed

Cause of late STILLBIRTH

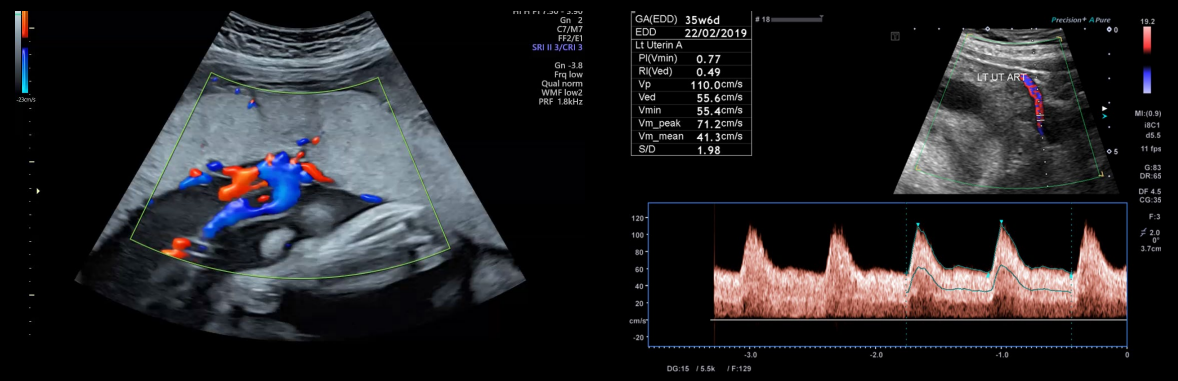
Placental Dysfunction

Anatomical

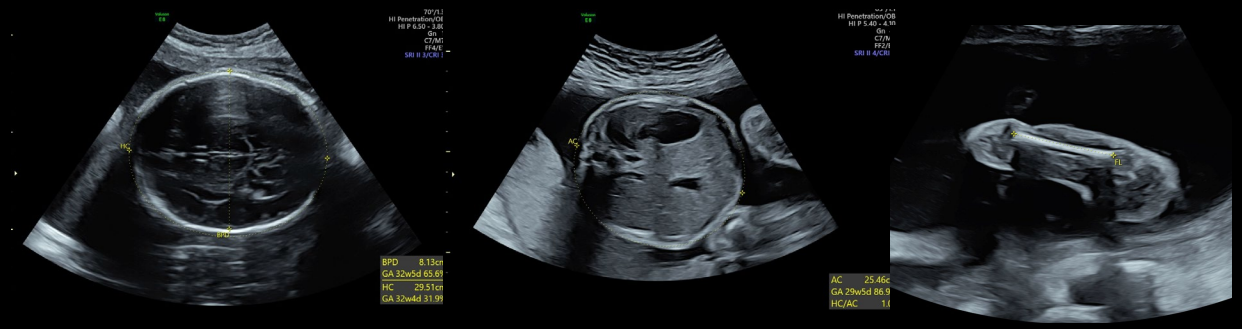


Haemodynamic

Resistance to flow

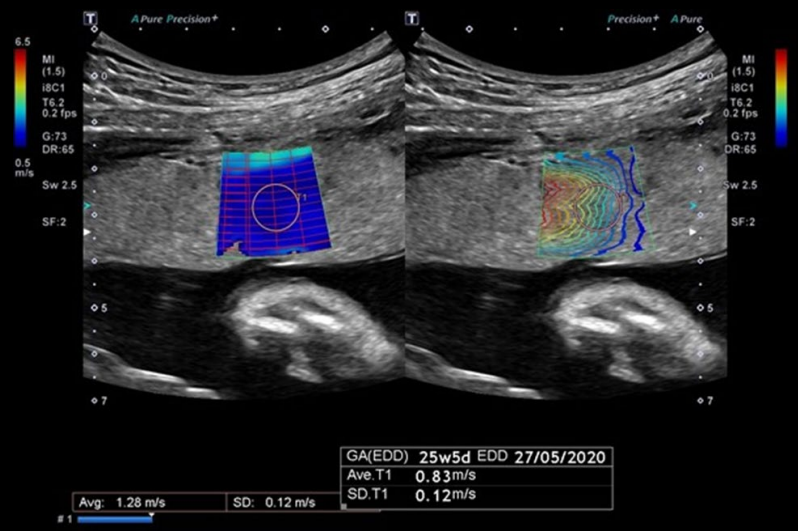


Biometric

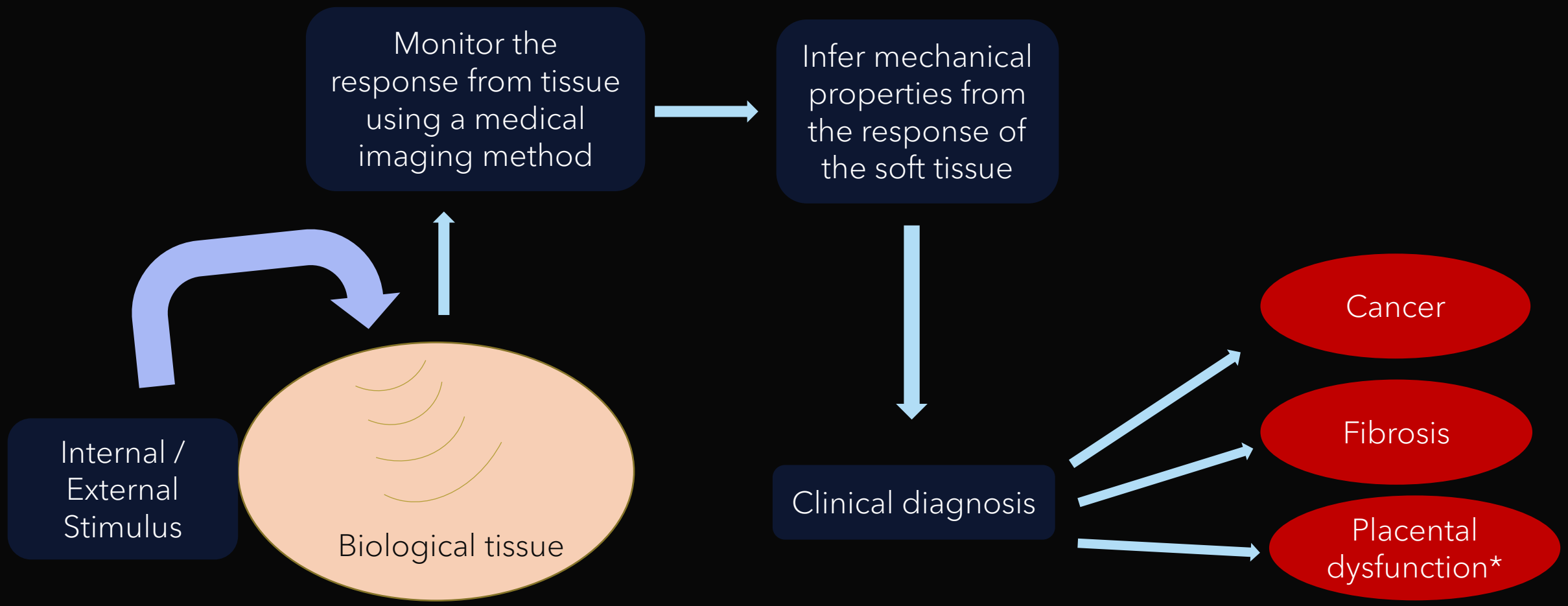


BPD > HC > AC > FL - EFW

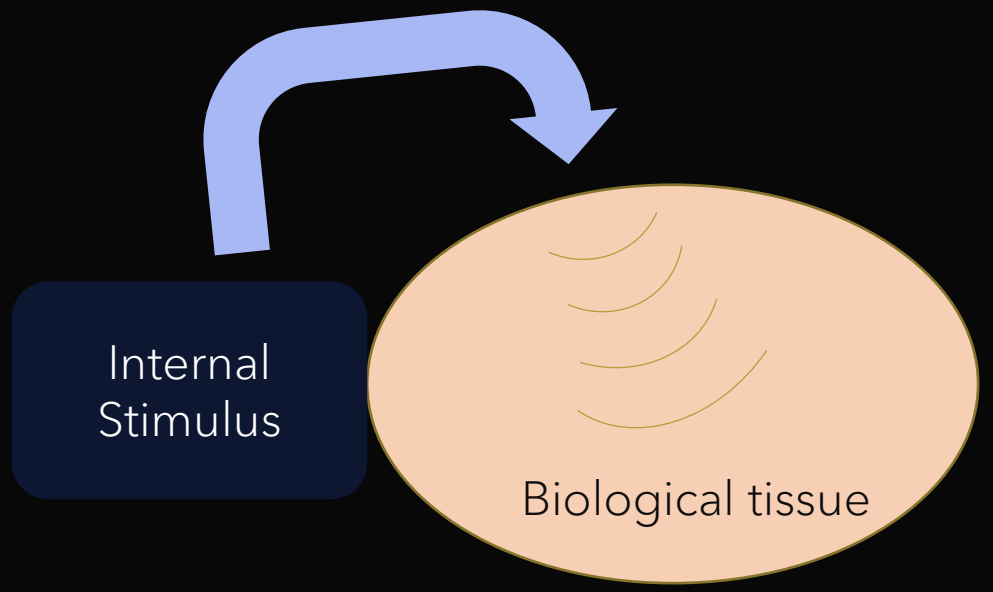
Elastography

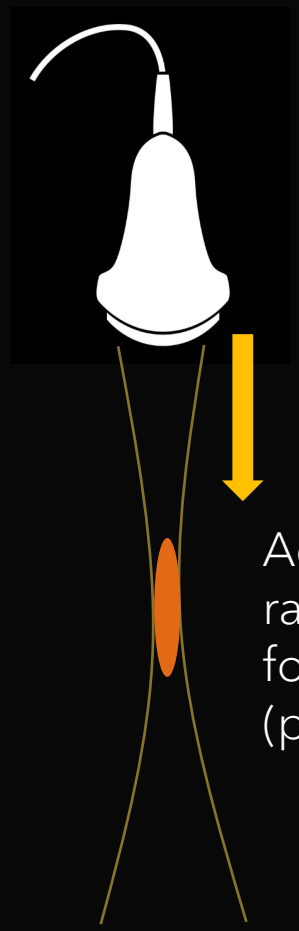


Elastography

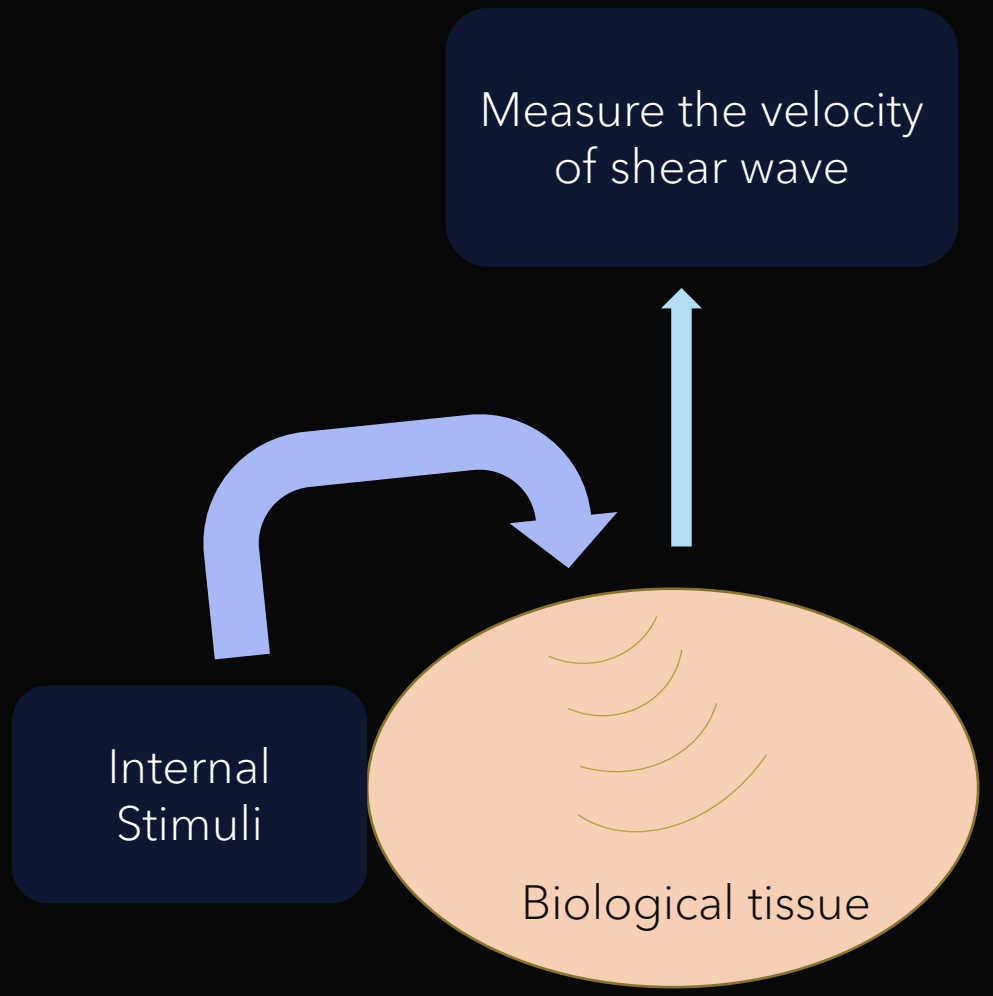


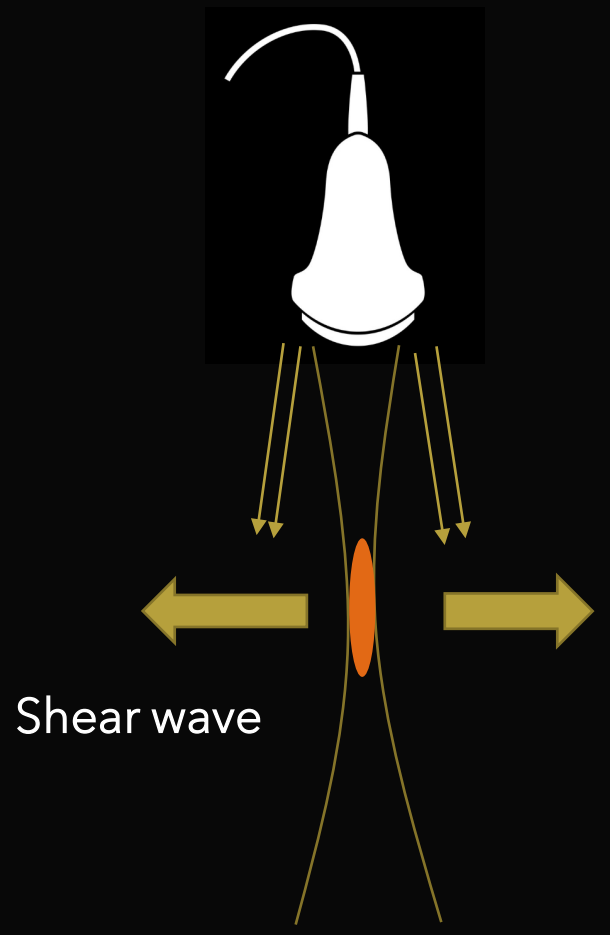
Shear Wave Elastography (SWE)





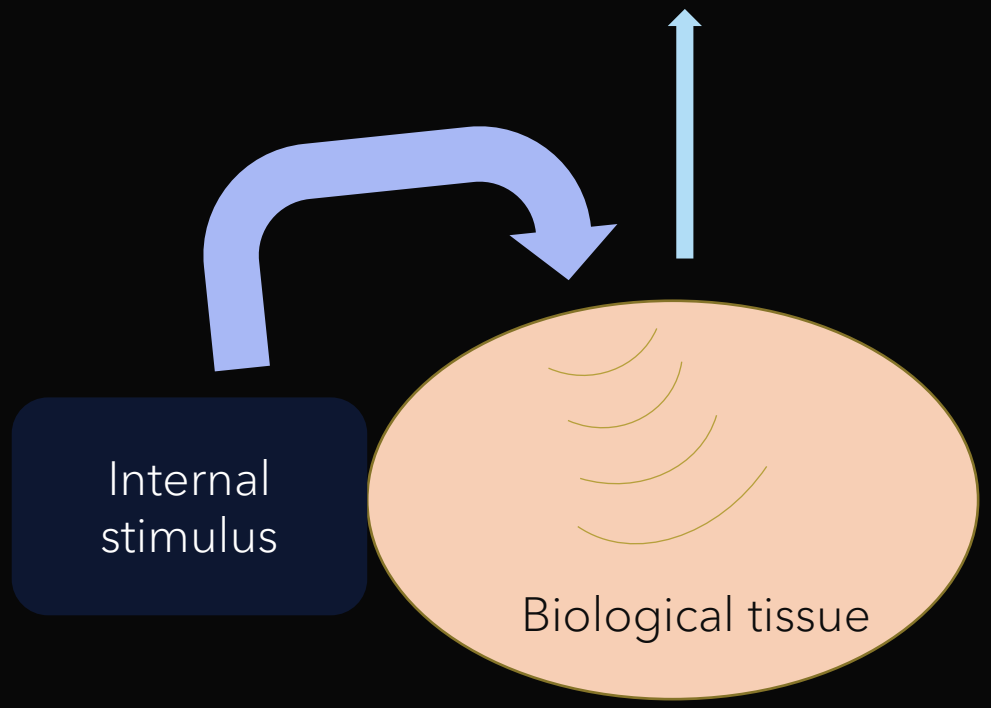
Shear Wave Elastography (SWE)

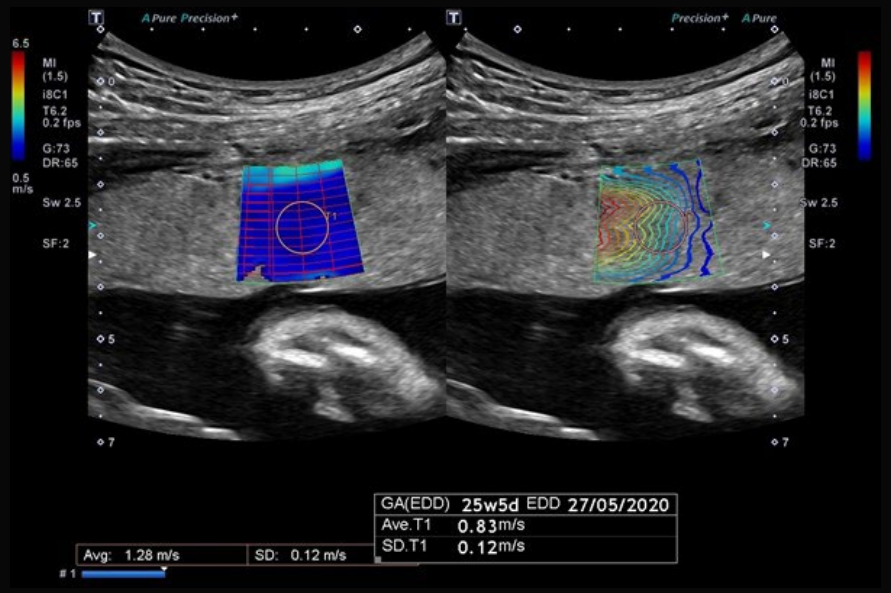




Shear Wave Elastography (SWE)

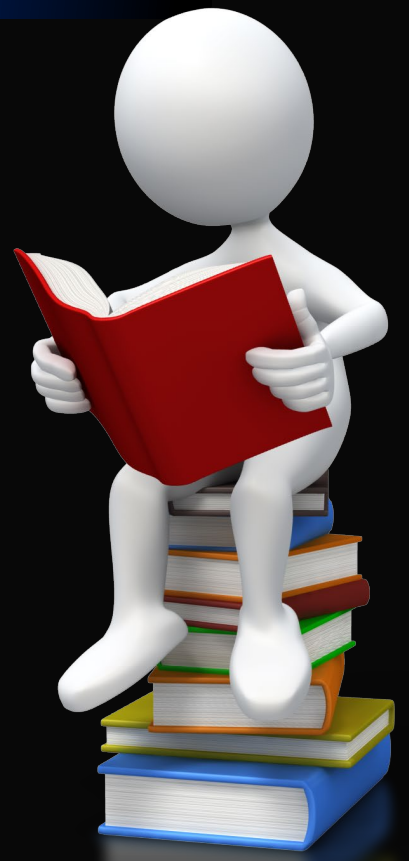
Measure the velocity of shear wave → Shear wave velocity (m/s) ∝ tissue stiffness





To investigate the application of SWE to human placenta tissue.

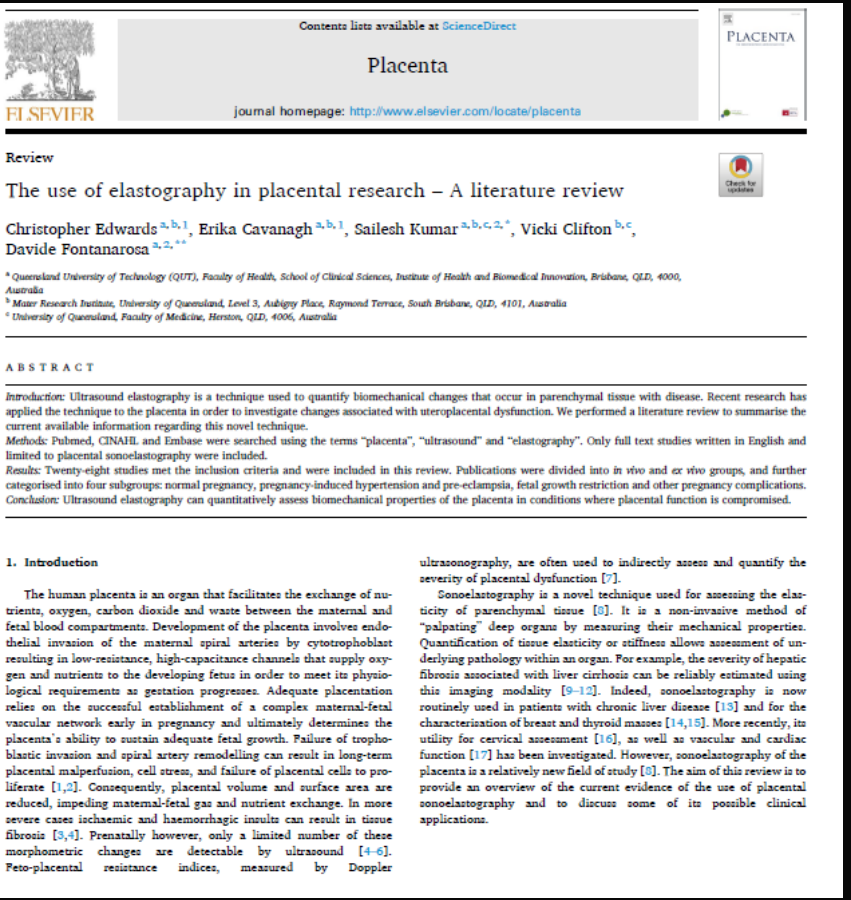
1. Understand how best to apply modern SWE technology to placental tissue
2. Longitudinally investigate the biological predictors of in vivo placental SWV measurements when acquired in normal pregnancy.
3. Identify the key factors (technical & physiological) to assist in the development of better diagnostic tools in the future.



Thesis – Chapter 1 & 2

Introduction and Literature Review

1. Association between increased SWV or stiffness (kPa) in abnormal placental tissue (pre-eclampsia , growth restriction) compared to normal controls
2. Measurements were system dependent and a range of ultrasound elastography techniques used
3. Most papers excluded imaging when the placenta was positioned posteriorly
4. Lack of knowledge related to the influence of gestational age and other maternal factors such as blood pressure and maternal weight status



Contents lists available at ScienceDirect

Placenta

journal homepage: http://www.elsevier.com/locate/placenta

Review

The use of elastography in placental research – A literature review

Christopher Edwards [a,b,1], Erika Cavanagh [a,b,1], Sailesh Kumar [a,b,c,2,*], Vicki Clifton [b,c], Davide Fontanarosa [a,2,**]

[a] Queensland University of Technology (QUT), Faculty of Health, School of Clinical Sciences, Institute of Health and Biomedical Innovation, Brisbane, QLD, 4000, Australia
[b] Mater Research Institute, University of Queensland, Level 3, Aubigny Place, Raymond Terrace, South Brisbane, QLD, 4101, Australia
[c] University of Queensland, Faculty of Medicine, Herston, QLD, 4006, Australia

ABSTRACT

Introduction: Ultrasound elastography is a technique used to quantify biomechanical changes that occur in parenchymal tissue with disease. Recent research has applied the technique to the placenta in order to investigate changes associated with uteroplacental dysfunction. We performed a literature review to summarise the current available information regarding this novel technique.
Methods: Pubmed, CINAHL and Embase were searched using the terms "placenta", "ultrasound" and "elastography". Only full text studies written in English and limited to placental sonoelastography were included.
Results: Twenty-eight studies met the inclusion criteria and were included in this review. Publications were divided into *in vivo* and *ex vivo* groups, and further categorised into four subgroups: normal pregnancy, pregnancy-induced hypertension and pre-eclampsia, fetal growth restriction and other pregnancy complications.
Conclusion: Ultrasound elastography can quantitatively assess biomechanical properties of the placenta in conditions where placental function is compromised.

1. Introduction

The human placenta is an organ that facilitates the exchange of nutrients, oxygen, carbon dioxide and waste between the maternal and fetal blood compartments. Development of the placenta involves endothelial invasion of the maternal spiral arteries by cytotrophoblast resulting in low-resistance, high-capacitance channels that supply oxygen and nutrients to the developing fetus in order to meet its physiological requirements as gestation progresses. Adequate placentation relies on the successful establishment of a complex maternal-fetal vascular network early in pregnancy and ultimately determines the placenta's ability to sustain adequate fetal growth. Failure of trophoblastic invasion and spiral artery remodelling can result in long-term placental malperfusion, cell stress, and failure of placental cells to proliferate [1,2]. Consequently, placental volume and surface area are reduced, impeding maternal-fetal gas and nutrient exchange. In more severe cases ischaemic and haemorrhagic insults can result in tissue fibrosis [3,4]. Prenatally however, only a limited number of these morphometric changes are detectable by ultrasound [4–6]. Feto-placental resistance indices, measured by Doppler ultrasonography, are often used to indirectly assess and quantify the severity of placental dysfunction [7].

Sonoelastography is a novel technique used for assessing the elasticity of parenchymal tissue [8]. It is a non-invasive method of "palpating" deep organs by measuring their mechanical properties. Quantification of tissue elasticity or stiffness allows assessment of underlying pathology within an organ. For example, the severity of hepatic fibrosis associated with liver cirrhosis can be reliably estimated using this imaging modality [9–12]. Indeed, sonoelastography is now routinely used in patients with chronic liver disease [13] and for the characterization of breast and thyroid masses [14,15]. More recently, its utility for cervical assessment [16], as well as vascular and cardiac function [17] has been investigated. However, sonoelastography of the placenta is a relatively new field of study [8]. The aim of this review is to provide an overview of the current evidence of the use of placental sonoelastography and to discuss some of its possible clinical applications.

Edwards C, Cavanagh E, Kumar S, Clifton V, Fontanarosa D. The use of elastography in placental research – A literature review. Placenta 2020;99:78–88.
https://doi.org/10.1016/j.placenta.2020.07.014

Thesis – Chapter 3

Chapter 3 – Research Design

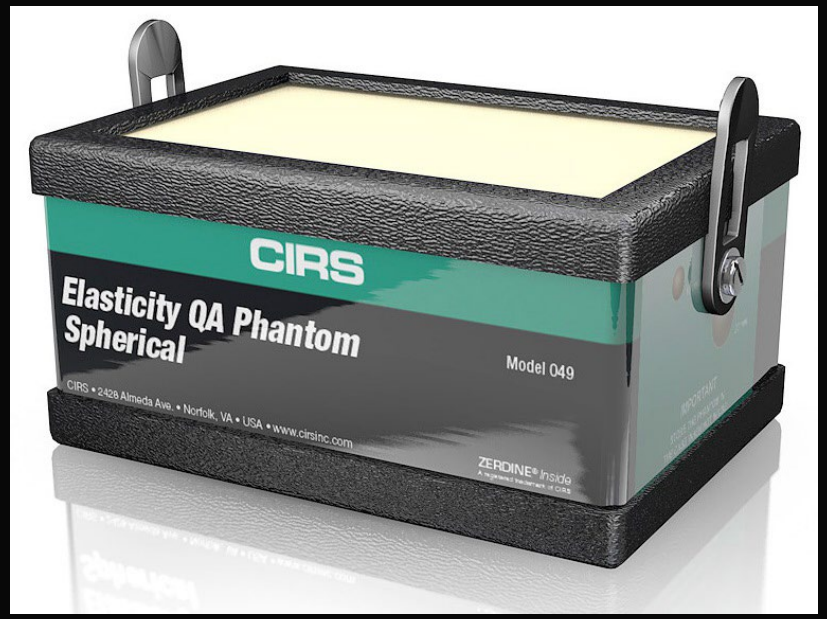
Laboratory phase

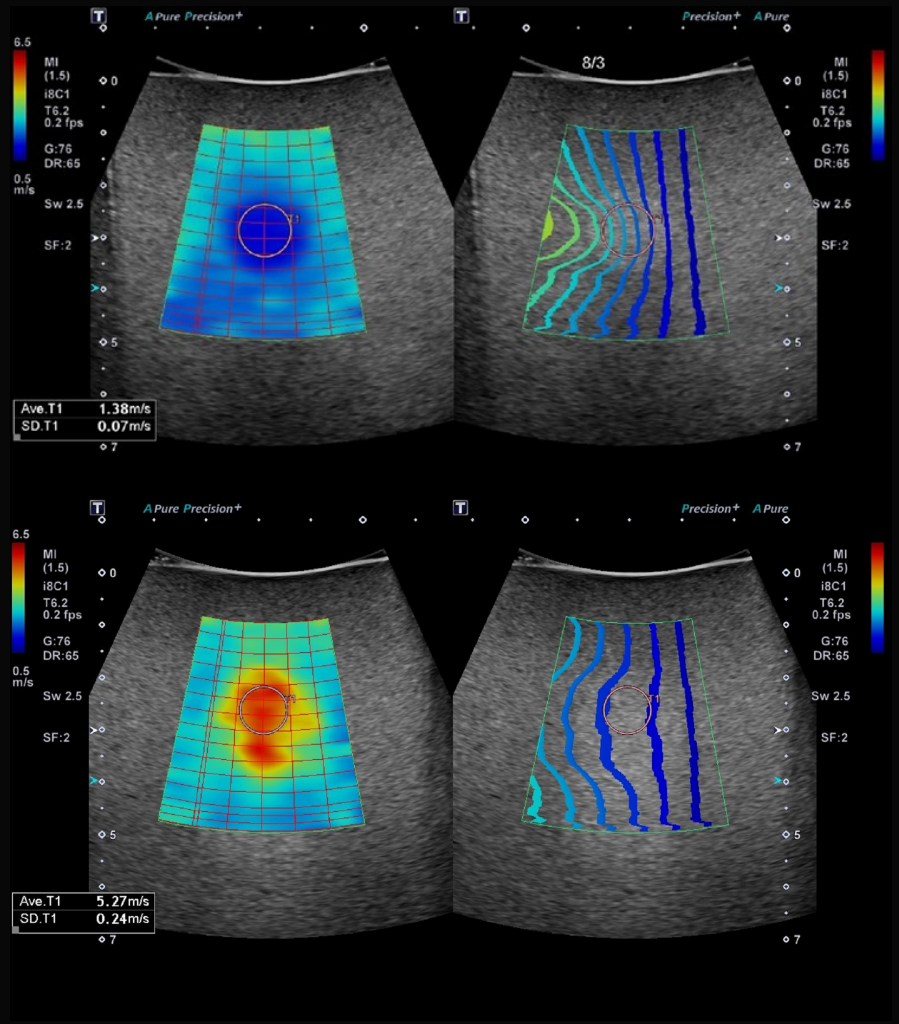
- Elastography QA phantom to understand system reliability

Clinical phase (QFC)

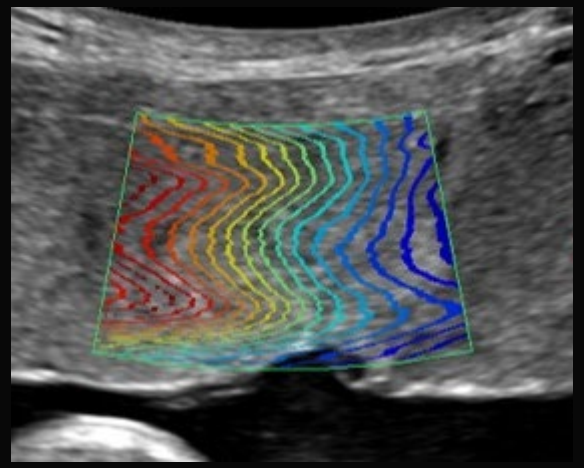
- Mater #HREC/16/MHS/113 / QUT #1800000995 ethics approval
- Prospective, observational, longitudinal study

Thesis – Chapter 4

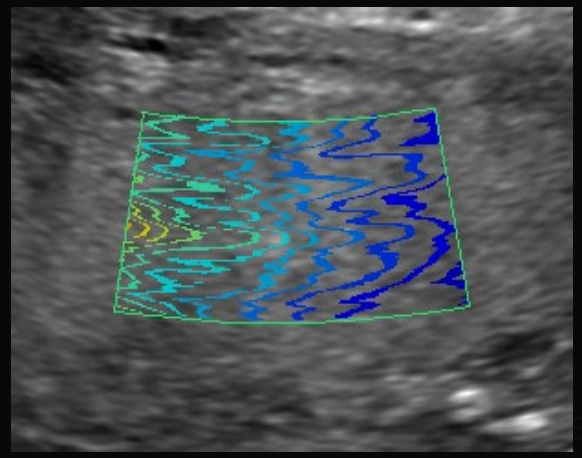




Shear Wave Elastography (SWE)



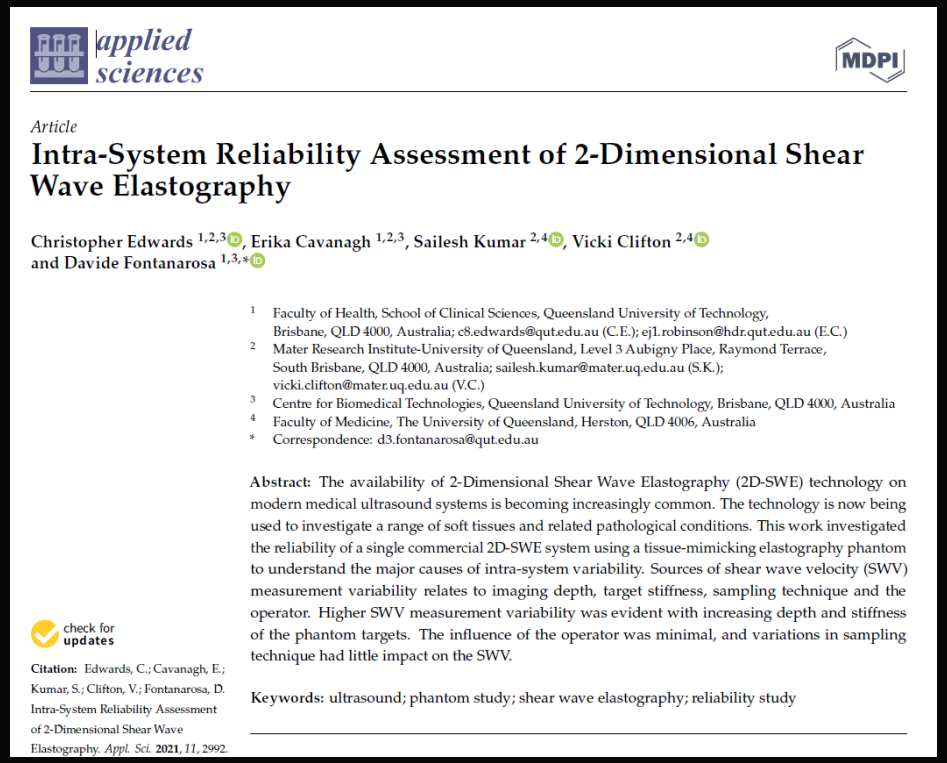
High reliability – lines are straight and parallel



Low reliability – Distorted lines and not parallel

Primary findings:

1. Good machine reliability of SWV measurements imaging soft (10kPa and 17kPa) phantom targets
2. Minimal operator variability even when we used an inexperienced operator.
3. No influence of measurement technique ROI (5mm vs 10mm)
4. Higher measurement variability with increasing depth and stiffness of the target



applied sciences
MDPI

Article

Intra-System Reliability Assessment of 2-Dimensional Shear Wave Elastography

Christopher Edwards [1,2,3], Erika Cavanagh [1,2,3], Sailesh Kumar [2,4], Vicki Clifton [2,4] and Davide Fontanarosa [1,3,*]

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2 Mater Research Institute-University of Queensland, Level 3 Aubigny Place, Raymond Terrace, South Brisbane, QLD 4000, Australia; sailesh.kumar@mater.uq.edu.au (S.K.); vicki.clifton@mater.uq.edu.au (V.C.)
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Abstract: The availability of 2-Dimensional Shear Wave Elastography (2D-SWE) technology on modern medical ultrasound systems is becoming increasingly common. The technology is now being used to investigate a range of soft tissues and related pathological conditions. This work investigated the reliability of a single commercial 2D-SWE system using a tissue-mimicking elastography phantom to understand the major causes of intra-system variability. Sources of shear wave velocity (SWV) measurement variability relates to imaging depth, target stiffness, sampling technique and the operator. Higher SWV measurement variability was evident with increasing depth and stiffness of the phantom targets. The influence of the operator was minimal, and variations in sampling technique had little impact on the SWV.

Keywords: ultrasound; phantom study; shear wave elastography; reliability study

check for updates

Citation: Edwards, C.; Cavanagh, E.; Kumar, S.; Clifton, V.; Fontanarosa, D. Intra-System Reliability Assessment of 2-Dimensional Shear Wave Elastography. *Appl. Sci.* **2021**, *11*, 2992.

Edwards, C., Cavanagh, E., Kumar, S., Clifton, V., & Fontanarosa, D. (2021). Intra-System Reliability Assessment of 2-Dimensional Shear Wave Elastography. Applied Sciences, 11(7), 2992. https://doi.org/10.3390/app11072992

Thesis – Chapter 5

Changes in placental elastography in the third trimester - Analysis using a linear mixed effect model

238 Patients
Data collected at 3 visits (24, 28 and 36 week)

Ultrasound Data

Gestational Age
Amniotic Fluid Index (AFI)
Doppler Indexes (MCA PI, UA PI, UtA PI)

Maternal Physiology

Body Mass Index (BMI)
Weight Difference
Blood pressure
Maternal Heart Rate

Elastography

Sampling depth

SWV

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Changes in placental elastography in the third trimester - Analysis using a linear mixed effect model

Christopher Edwards [a,b,g,*], Erika Cavanagh [a,b,g], Sailesh Kumar [a,b,c,**], Vicki L. Clifton [a,b], Danielle J. Borg [b,c], Jacob Priddle [e], Marie-Luise Wille [d,f,g], Christopher Drovandi [e], Davide Fontanarosa [a,g]

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[d] School of Mechanical, Medical, and Process Engineering, Faculty of Engineering, Queensland University of Technology, Brisbane, QLD, 4000, Australia
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ARTICLE INFO
Keywords:
Placenta
In vivo
Shear-wave elastography
Ultrasound
Placental elasticity
longitudinal study

ABSTRACT
Introduction: Research into the role of ultrasound elastography to assess compromised placental tissue is ongoing. There is particular interest in evaluating its potential in the investigation of changes associated with uteroplacental dysfunction. To date, there is limited data on how different maternal and fetal considerations, such as advancing gestational age, amniotic fluid Index (AFI) and maternal body mass index (BMI) may influence shear wave velocity (SWV) measurements. This study aimed to evaluate longitudinal changes in SWV throughout gestation and model these changes with other developing fetal and maternal physiological and biological characteristics.
Methods: The study utilised 238 singleton pregnancies and collected longitudinal data at repeated intervals in the 3rd trimester representing 629 individual data points. Linear mixed model regression analysis was used to identify significant predictors for SWV.
Results: From a total of ten variables selected for modelling, only gestational age, AFI, BMI, and sample depth were found to be significant predictors of placental SWV, and gestational age and AFI were found to have only a minimal impact on SWV.
Discussion: Sophisticated statistical modelling demonstrates that many of the expected maternal and fetal changes in the 3rd trimester have no or minimal impact on placental SWV. Understanding which factors influence placental SWV is essential to ascertain the technique's utility in managing pregnancies complicated by placental dysfunction in the future.

1. Introduction

Variations in placental "stiffness" are detectable via changes in shear wave velocity (SWV) measured with modern ultrasound shear wave elastography (SWE) techniques [1]. Increased placental SWV values compared to normal controls have been reported in women with pre-eclampsia [2–5], fetal growth restriction (FGR) [6,7], and those with gestational diabetes mellitus (GDM) [8]. These are all conditions where placental dysfunction is more prevalent and can lead to adverse maternal and perinatal outcomes, including stillbirth.

Identifying pregnancies at risk of these complications is a significant obstetric challenge [9]. Currently, serial biophysical assessment of fetal wellbeing using ultrasound to estimate fetal weight, assess liquor volume, and measure feto-placental vascular changes helps identify fetuses with overt growth restriction or those who are small for gestational age (GA). Prenatal identification of these infants is important as suboptimal

Edwards, C., Cavanagh, E., Kumar, S., Clifton, V. L., Borg, D. J., Priddle, J., Wille, M.-L., Drovandi, C., & Fontanarosa, D. (2021). Changes in placental elastography in the third trimester - Analysis using a linear mixed effect model. Placenta, 114, 83–89. https://doi.org/10.1016/j.placenta.2021.09.001

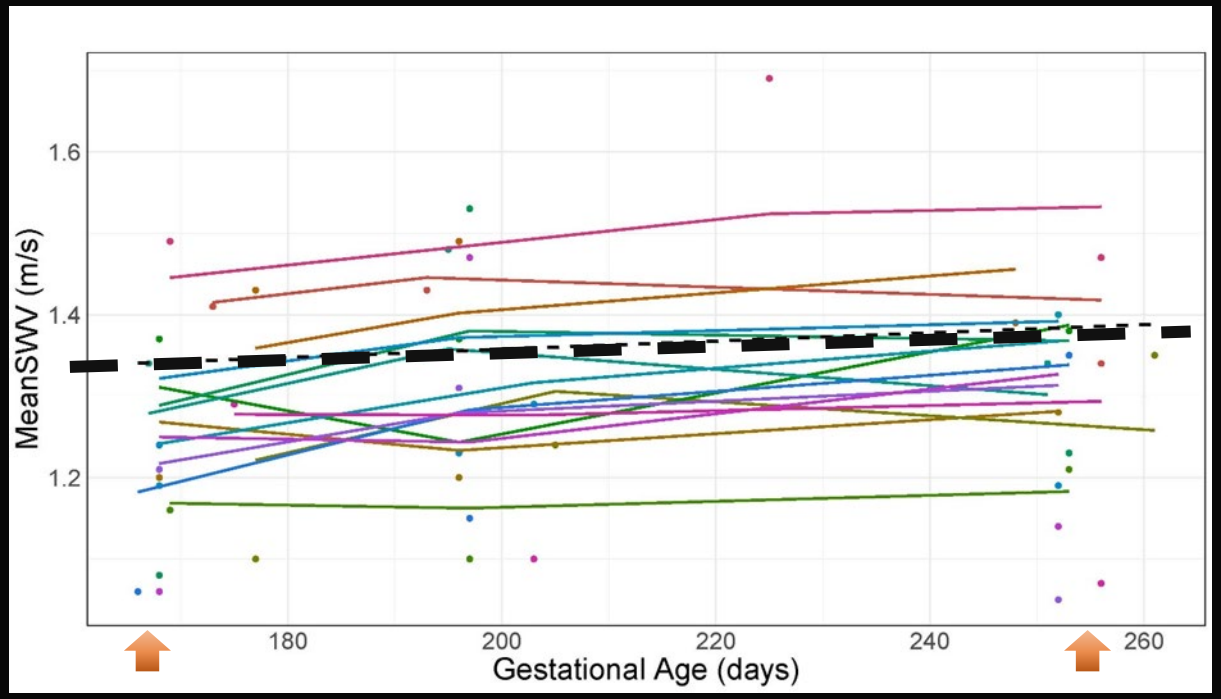
Primary findings:

Predictive model of SWV across the 3rd trimester.

$$Mean\ SWV_{ij} = \beta_0 + \beta_1 \times Gestational\ Age_{ij} + \beta_2 \times Sampling\ Depth_{ij} + \beta_3 \times BMI_{ij} + \beta_4 \times AFI_{ij} + \gamma_i + \varepsilon_{ij}$$

1. Largest fixed effects – Sampling depth and BMI
2. Minor impact - Gestational Age and AFI

Primary findings:



Fitted model scatterplot of MeanSWV vs Gestational Age of 15 random samples from the dataset.

Thesis – Chapter 6

Relationship between placental elastography, maternal pre-pregnancy body mass index and gestational weight gain

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Placenta

journal homepage: www.elsevier.com/locate/placenta

Relationship between placental elastography, maternal pre-pregnancy body mass index and gestational weight gain

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ABSTRACT

Introduction: Maternal obesity is a significant risk factor for poor pregnancy outcomes. Obesity is linked to abnormalities in placental tissue at term. The purpose of this study was to correlate changes in placental stiffness, measured via ultrasound elastography, with maternal pre-pregnancy body mass index and gestational weight gain.

Methods: Body Mass Index and gestation weight gain data was collected from 238 women. Elastography measurements were obtained via ultrasound at 24-, 28- and 36-weeks' gestation. An analysis using a linear mixed regression model assessed for the statistical significance of pre-pregnancy BMI, pregnancy weight gain and placental SWV (Shear Wave Velocity).

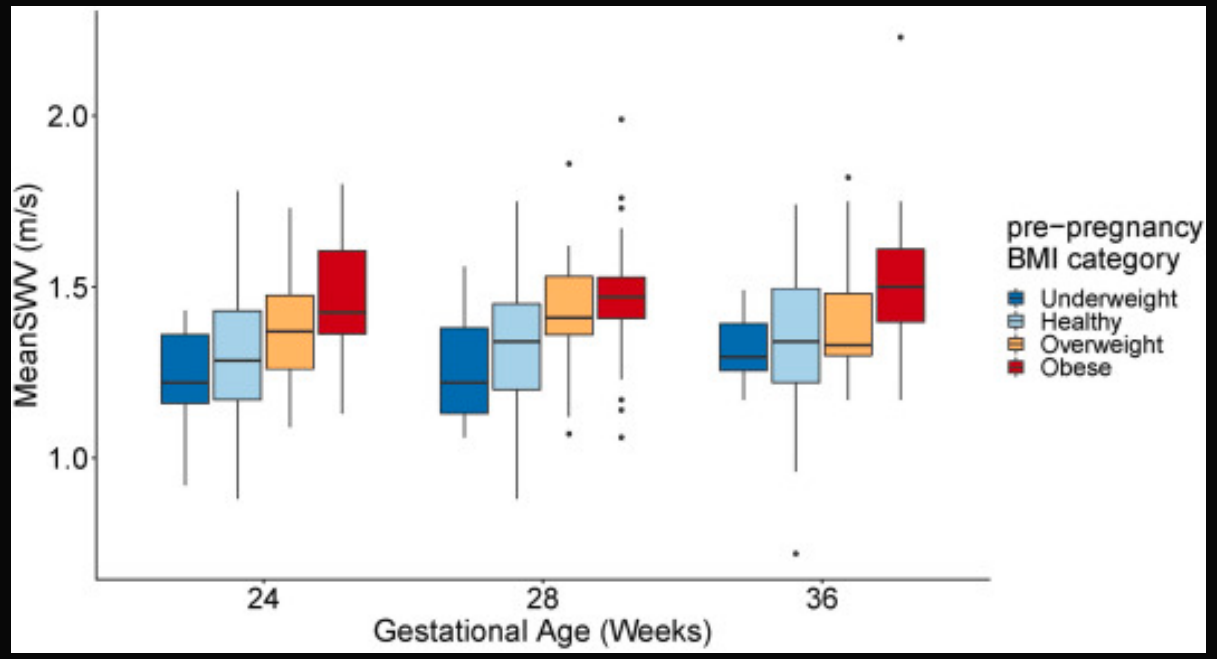
Results: Pre-pregnancy weight status has a significant impact on placental tissue stiffness detectable via ultrasound elastography. Placental tissue stiffness was highest in obese women, followed by overweight women. Obese women, on average, had a MeanSWV 0.11 m/s (95% CI (0.061–0.15) m/s, $p < 0.001$) above the healthy group throughout the 3rd trimester. Weight gain during pregnancy had a small impact on placental stiffness at the end of pregnancy. MeanSWV was 0.06 m/s (95% CI (0.03–0.10) m/s, $p < 0.001$) higher in the excessive weight gain group.

Discussion: Structural changes of the placenta detected via ultrasound elastography techniques are not exclusive to placental dysfunction conditions (pre-eclampsia and growth restriction) but are also associated with maternal obesity.

Edwards, C., Cavanagh, E., Kumar, S., Clifton, V. L., Borg, D. J., Priddle, J., Marie-Luise, W., Drovandi, C., & Fontanarosa, D. (2022). Relationship between placental elastography, maternal pre-pregnancy body mass index and gestational weight gain. Placenta, 121, 1–6. https://doi.org/10.1016/j.placenta.2022.02.015

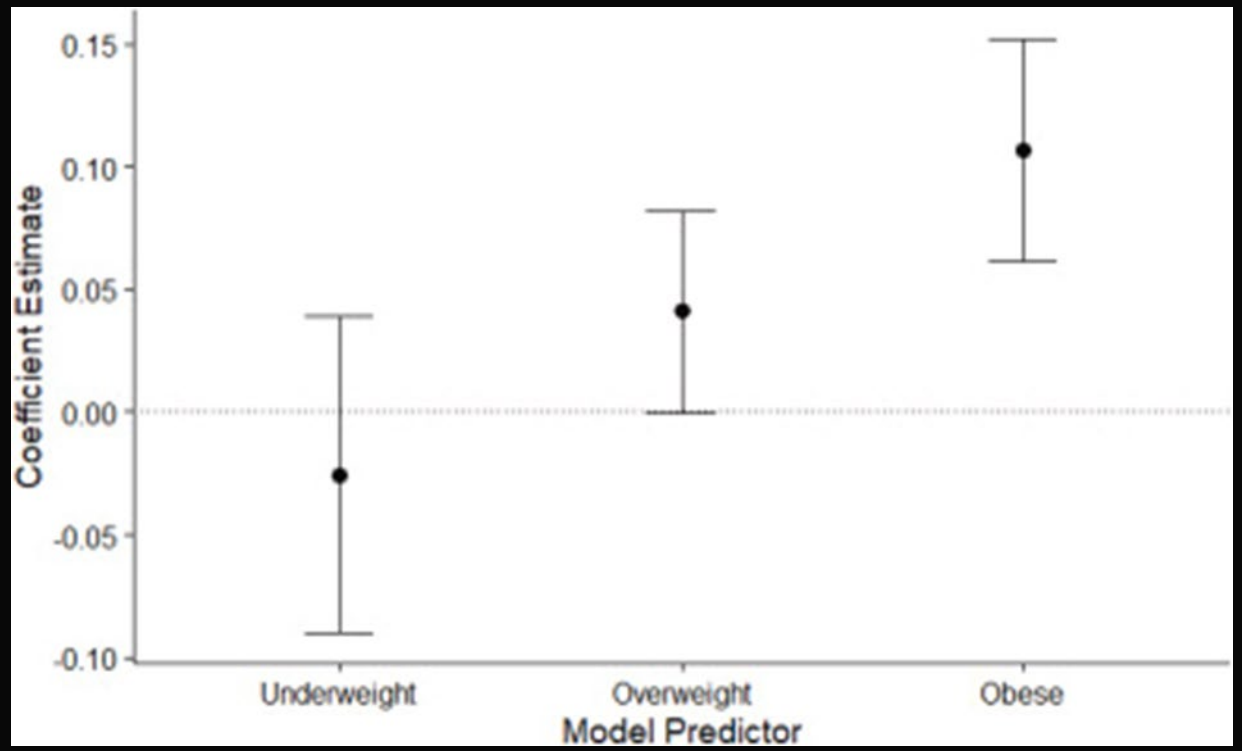
Primary findings:

1. Pre pregnancy weight status has a significant impact on SWV
   - Effect was seen at all time points (24, 28 and 36 weeks)
   - Independent of measurement conditions (sampling depth), and maternal factors (AFI and GA)



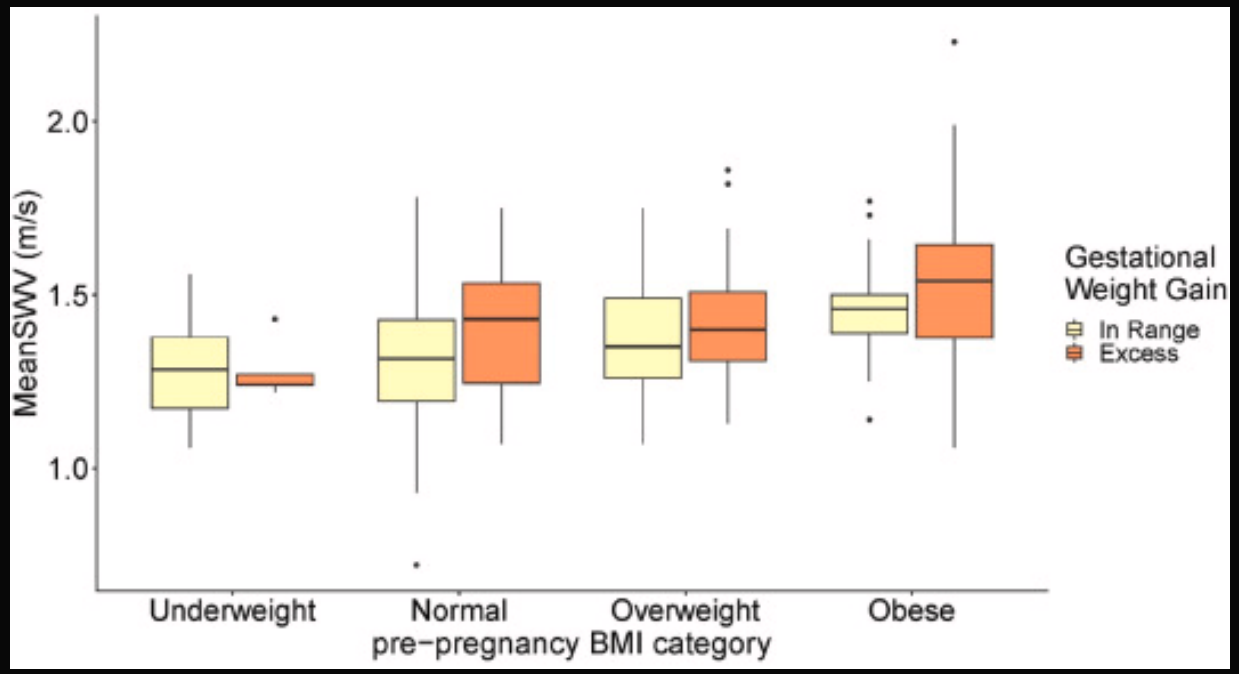
Primary findings:

1. Pre pregnancy weight status has a significant impact on SWV
   - Effect was seen at all time points (24, 28 and 36 weeks)
   - Independent of measurement conditions (sampling depth), and maternal factors (AFI and GA)
   - Progressive – Obese women showed greatest effect followed by the overweight group.



Primary findings:

2. Weight Gain
   - Small but significant effect on SWV
   - Pre-pregnancy weight status has the greatest impact.



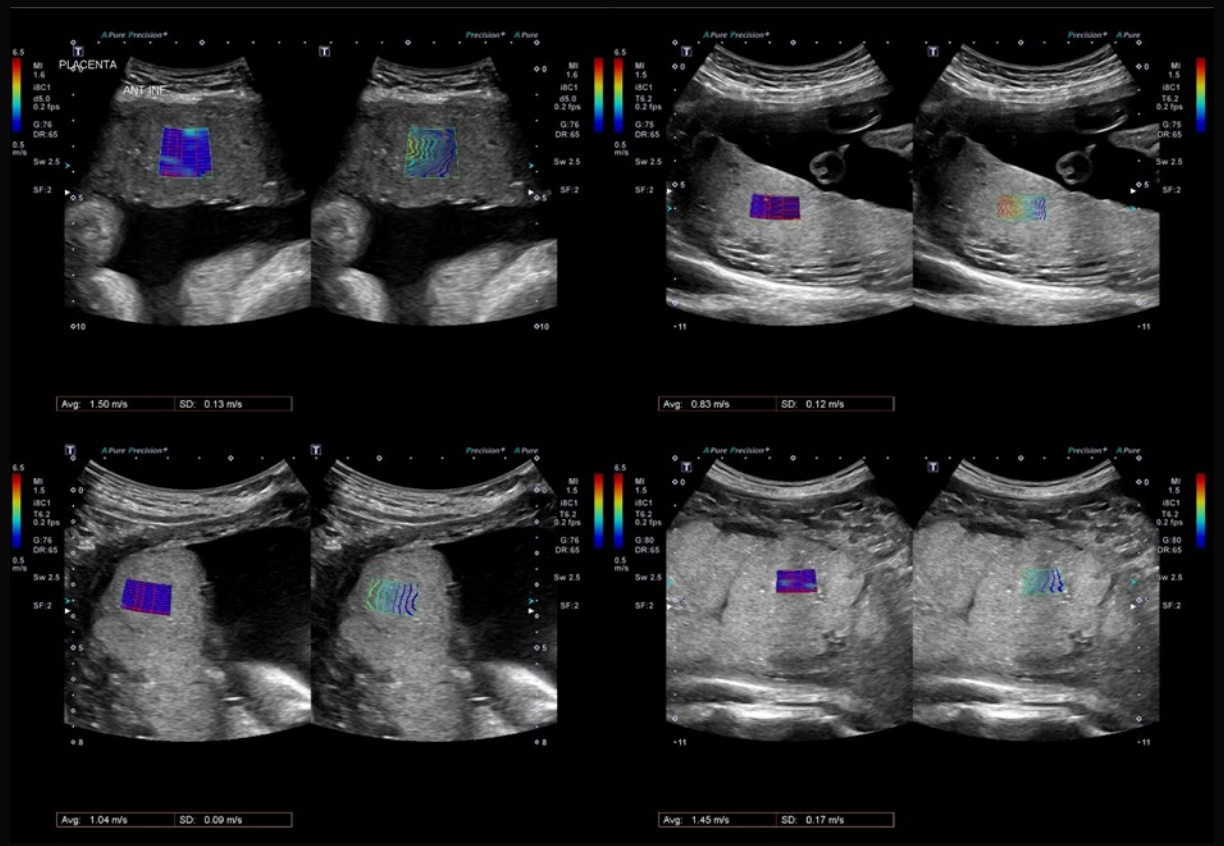
First non – invasive evidence of a structural impact of obesity on the placenta

Thesis – Chapter 7

Placental Position Does Not Hinder Measurement of Shear Wave Velocity

Under review

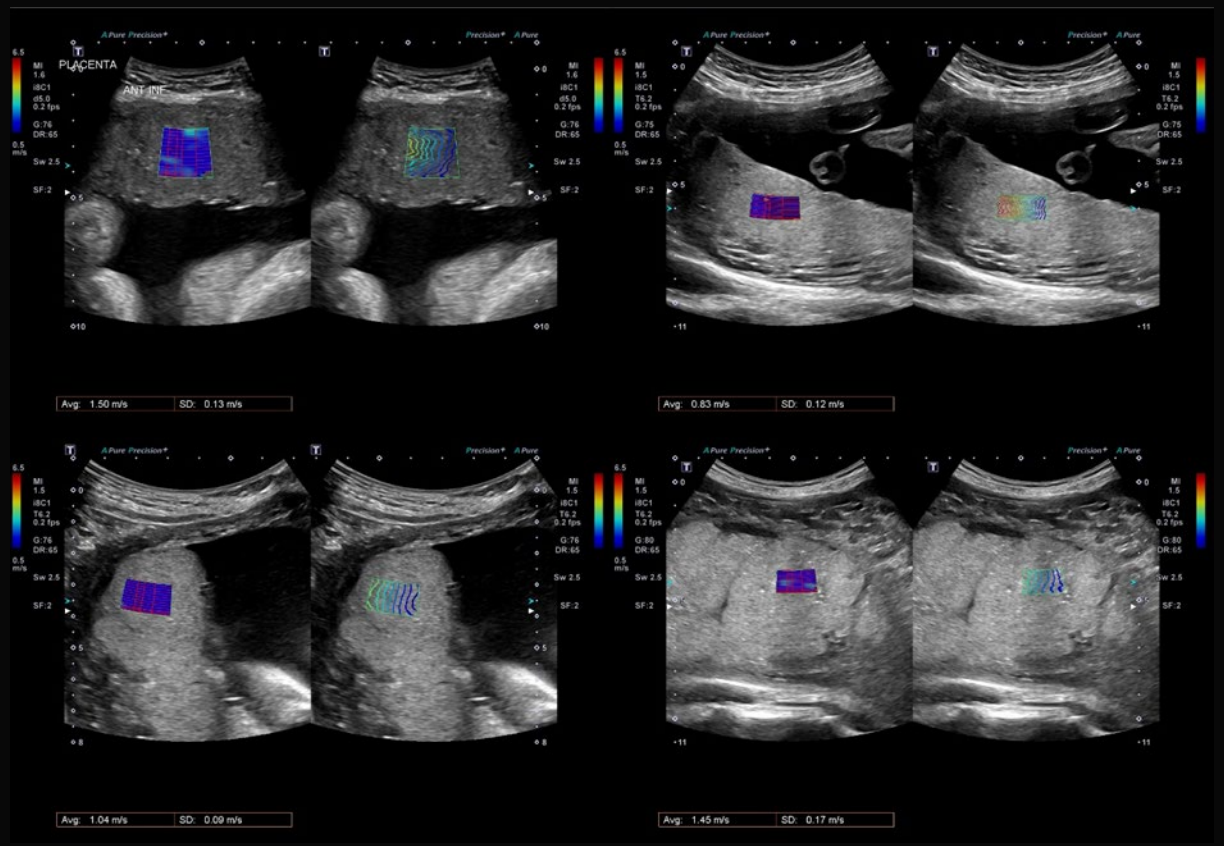
Ultrasound in Obstetrics and Gynecology



Thesis – Chapter 7

Primary findings:

1. Adjusting scan technique can result in reliable SWV measurements regardless of placental position.
2. Placental position is not a good predictor of SWV
3. Difference in SWV amongst individuals can be a function of sampling depth and not position.



Strengths / Limitations

- QA Phantom
- Model Analysis
- Cohort Study
- Experience of the Sonographers
- Pre-pregnancy weight status was self-reported

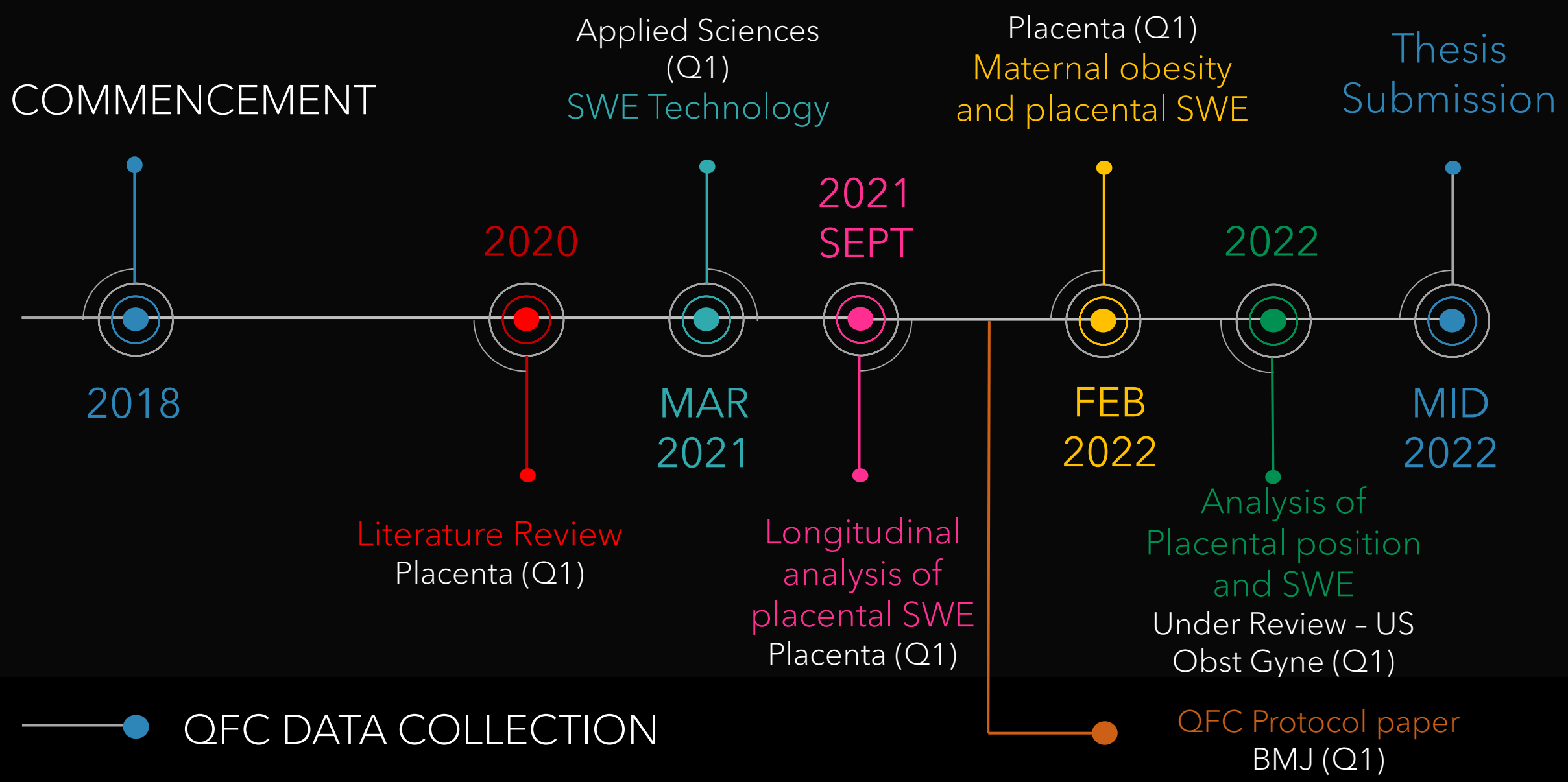
Key points

1. SWV is stable across late pregnancy
2. Technical consideration for sampling depth.
3. Linked maternal obesity and placental stiffness using ultrasound elastography.
4. Reliable measurements obtained from posterior placenta

Future development – research pathways

- Baseline model of normal placental biomechanical properties.
  - Tool to monitor interventions
  - Potential to develop better diagnostic tools .
    - combine sonoelastography with Biometry / Doppler / placental volume to make better predictions.
    - Utilise AI / Machine learning

Timeline


COMMENCEMENT
2018
2020
Literature Review
Placenta (Q1)
Applied Sciences
(Q1)
SWE Technology
MAR
2021
2021
SEPT
Longitudinal
analysis of
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MID
2022
QFC DATA COLLECTION
QFC Protocol paper
BMJ (Q1)

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